

# **Vivekanand Education Society's Institute of Technology**



## **Department of Computer Engineering**

### **Civility AI : Foul Language Detection System**

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## **Abstract**

The exponential growth of user-generated content on digital platforms has led to the proliferation of offensive and foul language, negatively impacting online civility. "Civility AI" is a proposed solution that leverages Natural Language Processing (NLP) and Machine Learning (ML) techniques to detect and classify offensive language in real-time. This synopsis reviews the methodologies, tools, and evaluation strategies used in building the Civility AI system, referencing three key studies. Our model integrates traditional ML classifiers and word embedding strategies like TF-IDF, GloVe, and Word2Vec to ensure contextual understanding and robustness against code-mixed and multilingual input.

## **Introduction**

Social media, while offering opportunities for expression and connection, has become a platform for hate speech, cyberbullying, and abusive content. The nature of online discourse has changed dramatically in recent years, making platforms vulnerable to toxic interactions. These toxic communications can cause emotional harm, spread misinformation, and provoke real-world violence. Manual moderation is insufficient due to the sheer scale and dynamic nature of online communication. Moreover, such methods are expensive, inconsistent, and often too slow to respond to the real-time nature of offensive posts. Therefore, there is a pressing need for automated systems like Civility AI that can proactively identify and mitigate foul language using advanced computational techniques. The system is intended not just for social media platforms but can be extended to forums, gaming chats, educational platforms, corporate communication channels, and governmental initiatives to combat digital hate.

## **Problem Statement**

The challenge lies in detecting offensive language accurately and efficiently across diverse linguistic contexts, including code-mixed and multilingual texts. Offensive content can be subtle or implicit, making it difficult for rule-based systems to catch. Furthermore, online users often use creative spellings, special characters, or coded language to bypass filters. Traditional methods like blacklists and regex-based filters lack context awareness and scalability. These techniques often result in high false-positive rates, flagging non-offensive content or failing to detect cleverly disguised slurs.

Additionally, there is a lack of publicly available annotated datasets that are culturally sensitive and linguistically diverse. A further issue is the generalization of models across domains—language that is considered foul in one platform (e.g., gaming) may be casual in another (e.g., comedy forums).

This project aims to develop a robust, scalable model that can detect, classify, and analyze foul language with high precision and minimal false positives. It also aims to adapt to evolving language patterns and slang, ensuring longevity and relevance.

## Literature Survey

| Paper                                                                                                        | Author Name                                                                                                                   | Key Takeaways                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Offensive Language Detection on Social Media using Machine Learning                                          | Rustam Abdrakhmanov , Serik Muktarovich Kenesbayev , Kamalbek Berkimbayev, Gumyrbek Toikenov , Elmira Abdrashova , Oichagul A | Deep Learning Models Outperform Traditional ML in Offensive Language Detection<br>Across studies, advanced architectures—particularly BiLSTM and BiRNN—consistently outperform classic ML methods (e.g., SVM, Naive Bayes, Logistic Regression) in capturing contextual nuances of hate speech and offensive content, thanks to their ability to model long-range dependencies and bidirectional context. |
| Deep learning based sentiment analysis and offensive language identification on multilingual code-mixed data | Kogilavani Shanmugavadivel , V. E. Sathishkumar                                                                               | Code-Mixed and Multilingual Data Pose Unique Challenges<br>Offensive language and sentiment detection in code-mixed datasets (e.g., Tamil-English, Malayalam-English) is harder due to informal writing, transliteration, and noise. Pretrained transformer models like BERT, RoBERTa, and Adapter-BERT with specialized embeddings (e.g., GloVe) improve accuracy for such cases.                        |
| ABUSIVE LANGUAGE DETECTION USING NLP                                                                         | Kandarpa Venkata Abhiram, Panigrahi Srikanth                                                                                  | Feature Engineering Remains Relevant Alongside Deep Learning<br>Even with powerful DL models, feature engineering—such as character n-grams, linguistic features are still useful for improving abusive language detection, especially in noisy social media environments.                                                                                                                                |
| Hateful Symbols or Hateful People? Predictive Features for Hate Speech Detection on Twitter.                 | Waseem, Z., & Hovy, D.                                                                                                        | Evaluation Highlights Trade-Off Between Precision and Recall<br>All three works stress balancing false positives and false negatives. Misclassifying benign content as offensive risks censorship, while missing offensive content allows harm.                                                                                                                                                           |
| Large Scale Crowdsourcing and Characterization of Twitter Abusive Behavior.                                  | Founta, A. M., Djouvas, C., Chatzakou, D., Leontiadis, I.                                                                     | Ethical, Scalability, and Interpretability Issues Persist<br>Despite technical gains, challenges remain: dataset bias, cultural and language variation, computational costs for large-scale deployment, and the “black box” nature of deep models hinder transparency. Human-in-the-loop moderation and interpretable AI methods are proposed to ensure fairness and adaptability.                        |

## Objectives and Methodology

The primary objective of **Civility AI** is to develop a robust, scalable, and context-aware foul language detection system capable of operating across diverse linguistic and cultural contexts. The system aims to accurately identify offensive or abusive content in real time while minimizing false positives and adapting to evolving patterns of online communication. Specific goals include:

- **Accurate Detection:** Achieve high precision and recall in identifying offensive and foul language.
- **Multilingual and Code-Mixed Support:** Handle regional languages and mixed-language text effectively.
- **Context Awareness:** Differentiate between genuinely offensive expressions and benign language in casual or domain-specific contexts.
- **Low False Positive Rate:** Ensure trustworthy performance without over-flagging.
- **Adaptability:** Update detection capabilities to incorporate new slang, spelling variations, and evolving offensive terms.
- **Scalable Deployment:** Enable integration across multiple platforms such as social media, gaming chats, educational portals, and corporate communication tools.

**To achieve these objectives, the following methodology will be employed:**

1. Data Collection and Preprocessing

Publicly available datasets such as the *Hate Speech and Offensive Language Dataset*, *OLID*, and *FIRE 2021 Dravidian CodeMix* will be used, along with custom-collected data from sources like Twitter and Reddit. Preprocessing will involve noise removal (URLs, emojis, punctuation), text normalization (lowercasing, tokenization, stopwords removal), deduplication, and dataset balancing through over-sampling or under-sampling.

2. Feature Representation

Text will be transformed using TF-IDF for frequency-based features, while semantic understanding will be captured through GloVe and Word2Vec embeddings. A curated offensive lexicon will be integrated to improve coverage and detection accuracy.

3. Model Development and Training

Traditional machine learning models such as Logistic Regression, Naive Bayes, and Support Vector Machines (SVM) will be trained and compared. The modular architecture will allow easy integration of improved algorithms in the future.

4. Multilingual and Code-Mixed Handling

Language detection will be performed on each text input, followed by preprocessing and embedding strategies tailored for multilingual and mixed-language scenarios.

5. Evaluation and Optimization

Model performance will be assessed using Accuracy, Precision, Recall, **F1-score**, **ROC-AUC**, **Kappa Score**, and **Matthews Correlation Coefficient**. K-fold cross-validation will ensure generalization, while tools like LIME and SHAP will provide explainability and bias detection.

6. Deployment and Feedback Integration

The system will be deployed as an API using Flask or FastAPI for seamless integration with external platforms. An optional moderator interface will allow human review of flagged content, enabling a semi-supervised feedback loop for continuous improvement.

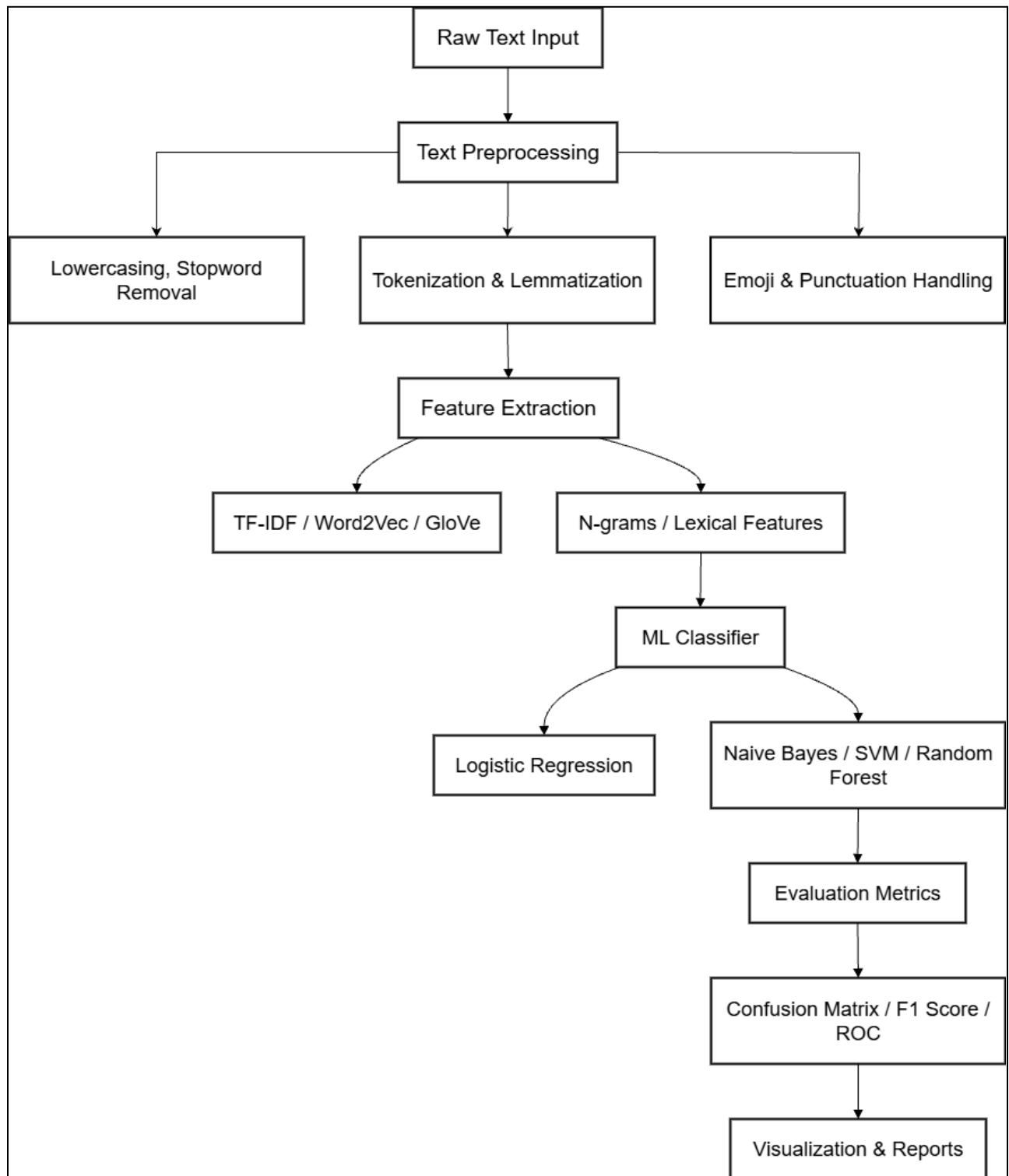
## Proposed Solution

1. Text preprocessing and cleaning techniques to prepare input data by removing noise such as URLs, emojis, and punctuation, and converting text to lowercase.
2. Traditional ML models such as Logistic Regression, Naive Bayes, and Support Vector Machines (SVM), each selected for their performance on text classification tasks.
3. Word embeddings using TF-IDF for term frequency-based scoring, GloVe and Word2Vec for semantic similarity and context-aware vectorization.
4. Multilingual and code-mixed text support to accommodate posts that mix languages or dialects, which are especially common in regional or bilingual communities.
5. Custom Lexicon Matching to enhance detection using curated lists of offensive terms.

The system pipeline includes preprocessing, embedding, classification, evaluation, and deployment modules. This modular approach ensures flexibility, allowing the integration of new algorithms or datasets with minimal disruption.

A user interface may also be developed for content moderators to interact with flagged data and provide human feedback, contributing to a semi-supervised improvement loop.

## Block Diagram :





# Hardware , Software and tools Requirements

## 1. Software:

- Python 3.8+ (with pip)
- Libraries:
  - scikit-learn for modeling
  - nltk and spaCy for preprocessing
  - gensim for embeddings
  - matplotlib and seaborn for visualizations
  - Flask or FastAPI for API deployment

## 2. Tools:

- Development: Jupyter Notebook, VS Code
- Deployment: Google Colab, AWS EC2, Heroku

## 3. Datasets and Preprocessing Details:

- Hate Speech and Offensive Language Dataset
- OLID (Offensive Language Identification Dataset)
- FIRE 2021 Dravidian CodeMix

Custom datasets may also be collected via Reddit or Twitter APIs using Python scripts. Preprocessing these requires additional steps:

- Deduplication
- Balancing (under-/over-sampling)
- Manual or semi-automated annotation

## Proposed Evaluation Measures

To validate Civility AI's performance, we consider:

- **Accuracy:** Overall success rate
- **Precision:** Correctly flagged offensive content vs. all flagged
- **Recall:** How many true offenses were actually detected
- **F1-Score:** Balances precision and recall
- **ROC-AUC:** Measures separability
- **False Positive/Negative Rates:** To evaluate trustworthiness

### Advanced Measures:

- **Kappa Score:** Agreement measure beyond chance
- **Matthews Correlation Coefficient:** For imbalanced datasets
- **Model Explainability Tools:** LIME or SHAP for transparency

All metrics will be calculated for each model and compared side-by-side. Charts will be provided for comparative analysis.

## Conclusion

Civility AI is a scalable, adaptable, and interpretable solution to detect and classify foul and offensive language in online content. Designed around traditional NLP and ML tools, it offers a transparent and low-resource alternative to deep learning models, without sacrificing performance. The system is robust across domains and is extendable to various platforms with minimal tuning. With ongoing improvements in language modeling and real-time deployment options, Civility AI represents a crucial advancement in fostering respectful online interaction.

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